
A Characterization of Steerability Limits in Frozen 3D Molecular Priors

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Abstract

1 Steering a frozen generative model toward an objective assumes the model can
2 be moved without retraining. We test this on Quetzal, an autoregressive 3D
3 molecular model trained on GEOM-Drugs, using 51 configurations spanning three
4 guide architectures, four training objectives and four fine-tuning scopes. All reach
5 approximately the score obtained by drawing the same number of molecules from
6 GEOM-Drugs itself, and on Perindopril MPO none exceeds it. The generated
7 molecules are nonetheless valid, unique and novel relative to the training corpus
8 (novelty > 0.9 , mean nearest-neighbour similarity ≈ 0.42), so the bound applies
9 to the achievable score rather than to the chemistry explored. We attribute it to the
10 confidence of the converged prior: 72% of atom decisions have a top-1 logit margin
11 above 8, and this confidence increases along the construction path, with guides
12 changing the sampled atom in up to 89% of states at the first position but under
13 10% by the fifth. Fine-tuning the prior’s own weights does not overcome the bound;
14 full-weight fine-tuning is the lowest-scoring configuration we test. We conclude
15 that the achievable objective value depends primarily on the pretraining distribution,
16 and that goal-directed scores are difficult to interpret without a matched-budget
17 dataset baseline.

18 1 Introduction

19 Molecular design under a fixed oracle budget can be approached by search or by amortized sampling.
20 GFlowNets Bengio et al. [2021] belong to the latter, sampling objects with probability proportional
21 to a reward rather than seeking a single optimum, and assigning credit across whole construction
22 paths so that multiple high-reward modes are preserved. This makes them a reasonable choice for
23 tilting a frozen generative prior toward a property objective, in particular through relative trajectory
24 balance Venkatraman et al. [2024], which targets $p_{\text{prior}}(x)R(x)^\beta$ without retraining the prior. We
25 use a residual-RL setup Johannink et al. [2018] in which a small trainable guide modifies the output
26 of a large frozen model.

27 Across 51 configurations trained on a 3D molecular prior, varying the guide architecture, the training
28 objective and the intervention scope, the resulting scores fall within a narrow band around the score
29 the prior’s own training set achieves at the same 10k-sample budget (Figure 1). On Perindopril
30 MPO no configuration exceeds that baseline. The molecules produced are valid, unique and novel
31 with respect to the training set, so what is bounded is the achievable score rather than the region of
32 chemical space reached.

33 The pattern is consistent with the confidence of the converged prior, and that confidence is not uniform
34 along a trajectory. At the first atom the model conditions on nothing and a guide changes the sampled
35 atom in up to 89% of states; each subsequent atom adds context, the margin widens, and by the fifth
36 position the flip rate falls below 10%. Steering is therefore concentrated at the start of a construction

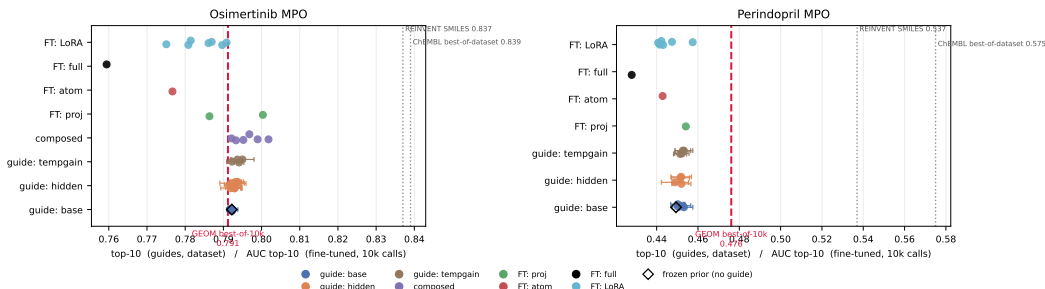


Figure 1: **Configurations cluster at the prior’s dataset score.** Each point is one configuration, coloured by family; the dashed red line is GEOM-Drugs’ best-of-10k baseline and dotted grey lines are published GuacaMol baselines. Across 30 (Osimertinib) and 21 (Perindopril) configurations the spread is 0.042 and 0.030, against gaps of 0.035 and 0.080 to the nearest published baseline. The metric is mixed: guides and the dataset are post-hoc i.i.d. samples without an oracle-call order, so we report final top-10, while fine-tuned runs record generation order and report AUC top-10 over 10k calls. Since top- k is non-decreasing in n , $\text{AUC top-10} \leq \text{top-10}$, so this axis slightly favours the guides, as does their smaller sample ($N=5000$ against $N=10,000$).

path, where the alternatives available are the prior’s own high-probability options. A natural objection is that the guides are too small, which we test by fine-tuning the prior’s own weights at three scopes and four LoRA ranks Hu et al. [2021]. Additional capacity does not help: performance declines as trainable parameters increase, and full-weight fine-tuning is the lowest-scoring configuration.

We make four contributions. First, we show that reward steering of a frozen molecular prior is bounded by its pretraining distribution across 51 configurations, while the molecules generated remain valid, unique and novel relative to that distribution. Second, we characterise the mechanism, reporting how guide influence decays along the construction path together with the accompanying margin growth. Third, we show that removing the capacity constraint reduces rather than improves steering, and quantify the associated validity–reward trade-off. Fourth, we argue that goal-directed scores are difficult to interpret without a matched-budget dataset baseline, and describe the protocol we use.

2 Related Work

3D molecular generation. Equivariant diffusion models generate atom types and coordinates jointly over a fixed point cloud Hoogetboom et al. [2022], with GEOM-Drugs Axelrod and Gomez-Bombarelli [2020] the standard corpus for drug-sized molecules. Quetzal Cheng et al. [2025] instead builds a molecule autoregressively, alternating a discrete atom-type decision with a coordinate diffusion conditioned on it. This factorisation is what makes logit-level intervention possible: the discrete decision is a categorical distribution over a small vocabulary and can be modified without altering the geometry model.

Steering pretrained models. Classifier and classifier-free guidance Dhariwal and Nichol [2021], Ho and Salimans [2022] steer diffusion models at sampling time, and controlled decoding and logit-bias methods serve the same purpose for autoregressive language models Yang and Klein [2021]. These methods assume a modest, well-directed perturbation of the prior’s output distribution suffices to reach the target region; we examine that assumption in the molecular setting. Reward fine-tuning offers the alternative of updating the model itself, via policy-gradient methods Black et al. [2023] or parameter-efficient adaptation such as LoRA Hu et al. [2021], while residual RL Johannink et al. [2018] keeps the model frozen and learns only a correction. We treat guides, LoRA adapters and full fine-tuning as points on a single capacity axis rather than as competing methods.

GFlowNets. GFlowNets Bengio et al. [2021] train a sampler whose terminal distribution is proportional to a reward, using trajectory-level credit assignment. Relative trajectory balance Venkattraman et al. [2024] adapts this to posterior sampling over a frozen prior and is the objective we

adopt. Replay buffers improve mode discovery under sparse rewards ?; we include one to separate exploration failure from the effect we report.

Benchmarking and oracle budgets. GuacaMol Brown et al. [2019] defines the goal-directed MPO tasks we use and reports a best-of-dataset baseline representing virtual screening, indicating the minimum a generative model must exceed to be doing more than retrieval. Practical Molecular Optimization Gao et al. [2022] adds a fixed oracle budget and reports the AUC of the top- k mean, rewarding sample efficiency rather than a single fortunate draw. Quality filters have shown that high MPO scores frequently co-occur with synthetically implausible molecules Brown et al. [2019]. The dataset baseline is defined in the original benchmark but often omitted in later work; Section 5 argues it is necessary for interpreting a reported score.

3 Background

Let p_{prior} be a pretrained generative model over molecules and $R(x) \in [0, 1]$ a black-box reward evaluated by an oracle. We wish to sample from the tilted distribution $p^*(x) \propto p_{\text{prior}}(x)R(x)^\beta$ under a fixed budget of oracle calls, without retraining p_{prior} . We use *steering* for any parameterised modification of the sampling distribution targeting p^* , whether an auxiliary module applied to a frozen prior (a *guide*) or an update of the prior’s own weights (*fine-tuning*).

Quetzal constructs a molecule as an interleaved sequence of atom types and coordinates. At step t , with prefix $s_{t-1} = (a_{<t}, x_{<t})$, a transformer trunk produces a hidden state h_t , a linear head emits atom-type logits $\ell_t = W_{\text{proj}}h_t$, and a second encoder produces a conditioner z_t for an 18-step coordinate diffusion:

$$p(\tau) = \prod_{t=1}^T \underbrace{p_{\text{atom}}(a_t | s_{t-1})}_{\text{categorical}} \underbrace{p_{\text{diff}}(x_t | z_t)}_{\text{diffusion}}. \quad (1)$$

The coordinate model is frozen in every experiment, and all interventions act on p_{atom} .

Relative trajectory balance requires a single learned scalar $\log Z_\phi$ to absorb the log-ratio for every trajectory,

$$\mathcal{L}_{\text{RTB}} = \mathbb{E}_{\tau \sim q_\theta} \left[\left(\log Z_\phi + \log q_\theta(\tau) - \log p_{\text{prior}}(\tau) - \beta \log R(x) \right)^2 \right], \quad (2)$$

whose minimiser is the tilted posterior. Detailed balance instead enforces local consistency between a learned state flow and the forward and backward policies, with the reward entering only at terminal states. Substituting (1) into (2), the ratio separates into an atom-type term and a coordinate term. When the trunk producing z_t is frozen, z_t is identical under q_θ and p_{prior} for any given prefix, the coordinate factors cancel exactly, and the atom-only log-ratio is not an approximation. When the trunk is trainable this no longer holds and the ratio incurs a bias bounded by the drift $\|z_t^\theta - z_t^0\|$, which we log (Appendix A.3); we therefore treat the head-only scope as the cleanest comparison.

GuacaMol benchmarks score each molecule with a benchmark-defined objective returning a value in $[0, 1]$, and report the mean of the top-1, top-10 and top-100 scores Brown et al. [2019]. Following PMO Gao et al. [2022] we fix an oracle budget of 10,000 molecules and report $\text{AUC}_k(B) = B^{-1} \int_0^B f_k(n) dn$, where $f_k(n)$ is the top- k mean among the first n calls. Since f_k is non-decreasing, $\text{AUC}_k(B) \leq f_k(B)$, and the difference reflects how quickly a method finds its best molecules.

4 Methodology

Intervention points. All methods modify only p_{atom} . Writing h for the trunk’s hidden state and W_{proj} for the frozen atom-type head, the three guide architectures are

$$\text{residual} : \tilde{\ell}(h) = W_{\text{proj}}h + g_\theta(h), \quad (3)$$

$$\text{temp/gain} : \tilde{\ell}(h) = W_{\text{proj}}h/T_\theta(h) + \gamma_\theta(h)g_\theta(h), \quad (4)$$

$$\text{hidden} : \tilde{\ell}(h) = W_{\text{proj}}(h + \delta_\theta(h)). \quad (5)$$

Table 1: Configurations evaluated. All share the frozen Quetzal prior and frozen coordinate diffusion.

Family	Intervention	Trainable	Variants
Frozen prior	—	0	—
Residual guide	output logits, (3)	MLP	$\{\text{DB, RTB}\} \times \{\text{replay on/off}\} \times \beta$
Temp/gain guide	logits + temperature, (4)	MLP + 2 heads	as above
Hidden guide	pre-projection state, (5)	MLP	as above
FT PROJ	W_{proj}	0.10 M	$\beta \in \{10, 20\}$
FT ATOM	$W_{\text{proj}} + \text{trunk}$	43 M	—
FT FULL	all weights	85 M	—
FT LoRA	W_{proj} , rank r	r -dependent	$r \in \{4, 16, 64\}$

Each is zero-initialised, so training begins at the frozen prior. Equation (3) is the direct residual-RL correction; (4) adds a per-state temperature intended to soften the prior where it is over-confident; (5) perturbs the input to the projection, so a small displacement is amplified by W_{proj} — a hidden delta of norm 0.1 produces a logit change of ≈ 46 , against ≈ 1.3 for an output residual of the same norm.

Removing the capacity constraint. We fine-tune the prior’s own weights under the same objective at three scopes: PROJ (the head only), ATOM (head and trunk) and FULL (all weights, including the coordinate denoiser). Only PROJ preserves the exact cancellation of Section 3. We also apply LoRA Hu et al. [2021] to the head at ranks $r \in \{4, 16, 64\}$; applied to W_{proj} this induces the perturbation $\delta\ell = (\alpha/r)BAh$, a rank- r linear guide, placing the guides and the fine-tuning scopes on a single capacity axis.

Measuring whether a guide changes anything. Aggregate reward differences do not distinguish a guide that is inert from one that acts but is mis-directed. We roll trajectories with the frozen prior and, at every state, compare the prior’s next-atom distribution with the guided one on the identical state, using a shared uniform draw so the two samplers are coupled. This gives *delivery*, the fraction of states at which the guided logits differ from the prior’s; the *sampled-flip rate*, the fraction at which the coupled draw selects a different atom; and the prior’s *top-1 margin* $\ell_{(1)} - \ell_{(2)}$, which indicates how large a residual must be to change the decision. Because the trajectories are the prior’s, every architecture is compared on the same state distribution.

Evaluation protocol. Every method is scored on the benchmark’s objective at a matched budget of 10,000 oracle calls, deduplicated on canonical SMILES. We apply the same budget to the dataset baseline, taking the best of 10,000 molecules drawn from GEOM-Drugs. Since Quetzal emits geometries, we also report the 3D-to-SMILES conversion rate, a failure mode the graph- and string-based baselines do not have. Alongside score we report uniqueness, internal diversity, Murcko scaffold diversity, compound-quality pass rate, and novelty and mean nearest-neighbour Tanimoto similarity against the training corpus, so that a score at the dataset baseline can be distinguished from a policy that has collapsed onto it.

5 Results

Configurations cluster at the dataset score. Figure 1 places all 30 Osimertinib and 21 Perindopril configurations on a single axis against the GEOM best-of-10k baseline. The spread across configurations is 0.042 and 0.030, while the gap to the nearest published baseline is 0.035 and 0.080: the differences between three guide architectures, four objectives and four fine-tuning scopes are smaller than the distance separating all of them from published methods. The frozen prior with no intervention scores 0.792 on Osimertinib against the dataset’s 0.791, and on Perindopril no configuration exceeds the dataset baseline. Table 8 gives the full breakdown.

The prior is confident, increasingly so along a trajectory. The guides are not inert. Delivery is 1.000 in every configuration, so the residual reaches the logits at every state, yet the coupled sampled-flip rate is below 1% overall. The prior’s mean top-1 margin is 16.4 and 72% of decisions exceed a margin of 8; within that high-margin regime the flip rate is at most 0.0006, against 0.017–0.033 on low-margin decisions, a difference that holds across architectures, objectives and rewards (Table 7).

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Figure 2: **Guide influence decays along the construction path.** Coupled sampled-flip rate against sequence position (left axis) with the prior’s mean top-1 logit margin on the same positions (right axis), averaged over configurations. Trajectories are rolled by the frozen prior, so all architectures are compared on an identical state distribution. The first atom is conditioned on nothing and is genuinely contestable; each subsequent atom narrows the choice, and the margin growth accounts for the decay.

146 Figure 2 resolves this along the trajectory: the flip rate reaches 0.89 at the first atom, falls below 0.10
147 by the fifth position and continues to decline, tracking the margin’s growth. Steering is therefore
148 concentrated at the start of a construction path, where the available alternatives are the prior’s own
149 high-probability options.

150 **The bound applies to score rather than to chemistry.** A score at the dataset baseline would be
151 unsurprising if the model were reproducing the dataset, but this is not what we observe. Across
152 fine-tuned configurations, novelty with respect to GEOM-Drugs is 0.86–0.97 and the mean nearest-
153 neighbour Tanimoto similarity to the training corpus is 0.40–0.47, so the molecules scored are not
154 close neighbours of training molecules. They are also well formed: 3D-to-SMILES conversion
155 holds at 0.93–0.95 against the frozen prior’s 0.926, uniqueness among valid molecules is ≥ 0.99 for
156 every configuration but one, internal diversity is 0.73–0.87, scaffold diversity 0.87–1.00, and quality
157 pass rates of 0.79–0.96 bracket the dataset’s 0.97. The guides show the same pattern from the other
158 direction: their distributional distance from the frozen prior is small ($FCD = 0.29$) while both sit at

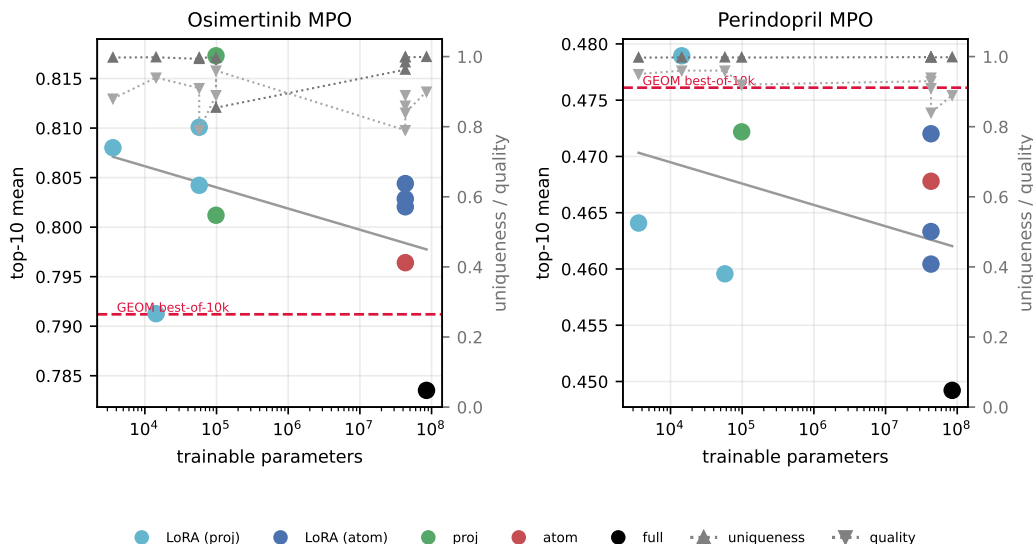


Figure 3: **Score declines as trainable capacity increases.** Benchmark score against the number of trainable parameters, from rank-4 LoRA through full-weight fine-tuning, with uniqueness and compound-quality pass rate on the secondary axis. Capacity is not the binding constraint: the configurations with the most parameters score lowest while shifting furthest from the prior’s distribution.

the same distance from the reference corpus (4.34 and 4.33). The models generate valid, diverse and novel molecules, and that exploration does not raise the score.

Additional capacity reduces steering, and buys distance at a cost. If the bound were set by guide capacity, fine-tuning the prior’s own weights should relax it. Figure 3 shows the opposite: performance declines as trainable parameters increase, and FULL fine-tuning, which has no capacity constraint, is the lowest-scoring configuration on both benchmarks, with the largest distributional shift and one of the lowest quality pass rates. The residual-scale sweep shows the same trade-off directly, with effect size growing up to 4× scaling and then reversing as validity falls (Appendix ??). The one configuration that clears the dataset baseline by a meaningful margin, PROJ at $\beta = 20$, does so while uniqueness falls from 0.997 to 0.854 and internal diversity to 0.73, the lowest we observe. Reward is obtained at the cost of validity, uniqueness and quality, and the exchange rate worsens with distance from the prior.

The result does not depend on the loss or on exploration. Detailed balance, relative trajectory balance and the KL variants agree to within seed noise, and a replay buffer improves mode discovery without changing the bound. Full numbers are given in Appendix D.

6 Discussion

A flat result is usually read as evidence that the method is too weak, with more capacity as the remedy. Our results do not support that reading: capacity is available and is used, since the residual reaches the logits at every state, and adding more of it lowers the score. The remaining explanation is a property of the prior. A converged autoregressive model expresses its training distribution through the confidence of its predictions, and that confidence accumulates along a trajectory, so an intervention on atom-type logits selects among the prior’s own modes at the start of construction and is carried by the prior thereafter. Reweighting a distribution can move probability mass within it but cannot easily move its optimum beyond its support.

If the achievable optimum depends on the prior’s training distribution, the choice of prior is a substantive design decision rather than a detail of the setup. GuacaMol’s objectives are defined over ChEMBL while our prior is trained on GEOM-Drugs, and GEOM’s best-of-10k falls below ChEMBL’s best-of-dataset on both benchmarks we study, by approximately the margin our methods

fail to close. The bound applies to score and not to novelty: our models reach chemistry the dataset does not contain, but chemistry that scores as the dataset does. This is why we regard the dataset baseline as necessary. It is defined in the original benchmark but often omitted, and without it at a matched budget a reported score is difficult to distinguish from what a virtual screen would produce. A related consequence is that reward gains here are confined to the region where outputs remain chemically sound, which limits the reward hacking that high MPO scores are known to admit Brown et al. [2019]; this follows from the same constraint that limits the reward rather than from any safeguard we designed.

Limitations. We freeze the coordinate diffusion throughout, so we characterise steering of the discrete decision only; a reward carried substantially by geometry would be outside the reach of every method we test, and we do not determine how much of these objectives is. We study a single prior, so the claim that the bound reflects the dataset rather than this particular model rests on ablations rather than a controlled substitution. Our fine-tuning runs are single-seed, with margins of the same order as the seed variance in the guide sweep. Finally, the temperature mechanism in (4) was inoperative: the learned T fell below 1 everywhere and was clamped, so those runs report a residual-plus-gain guide rather than the intended softening (Appendix F).

7 Conclusion

Across 51 configurations spanning three guide architectures, four training objectives and four fine-tuning scopes, all reach approximately the score the prior’s training set already achieves, while generating molecules that are valid, unique and novel with respect to that set. The bound applies to the objective value rather than to the chemistry reached, and is consistent with a converged prior whose predictive confidence grows along the construction path; removing the capacity constraint lowers the score rather than relaxing the bound.

The most direct test of this account is to substitute the prior: if the bound reflects the dataset, a model pretrained on a corpus aligned with the objective should reach optima this one cannot, and the achievable score should be predictable from the prior’s headroom on each benchmark. Priors with a more discrete construction process may also suit GFlowNet objectives better than one whose capacity lies largely in a frozen continuous component. Where gradients are least effective, at high-margin decisions, evolutionary or mixture-of-experts approaches may cover ground a single steered policy cannot. Underlying all of this is synthesizability, which this literature and our own work leave to a heuristic proxy.

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A Derivations

A.1 Notation

A trajectory $\tau = (s_0, s_1, \dots, s_T)$ is a sequence of partial molecules, where $s_t = (a_{\leq t}, x_{\leq t})$ contains the atom types and coordinates placed so far. Quetzal factorises the trajectory density as

$$p(\tau) = \prod_{t=1}^T p_{\text{atom}}(a_t \mid s_{t-1}) p_{\text{diff}}(x_t \mid z_t), \quad z_t = \text{enc}_2(a_{\leq t}, \text{enc}_1(s_{t-1})), \quad (6)$$

with $h_t = \text{enc}_1(s_{t-1})_{[-1]}$ the hidden state at the final position and $\ell_t = W_{\text{proj}} h_t$ the atom-type logits. We write q_θ for the steered policy and p_{prior} for the frozen model, and use superscripts θ and 0 to distinguish quantities computed under each.

A.2 Relative trajectory balance

RTB requires a single learned scalar $\log Z_\phi$ to absorb the log-ratio for every trajectory. Writing $\Delta(\tau) = \log q_\theta(\tau) - \log p_{\text{prior}}(\tau) - \beta \log R(x)$, the objective is

$$\mathcal{L}_{\text{RTB}} = \mathbb{E}_{\tau \sim q_\theta} \left[(\log Z_\phi + \Delta(\tau))^2 \right]. \quad (7)$$

At a zero of the residual, $\log q_\theta(\tau) = \log p_{\text{prior}}(\tau) + \beta \log R(x) - \log Z_\phi$ for all τ , so q_θ marginalises to $p^*(x) \propto p_{\text{prior}}(x) R(x)^\beta$ and Z_ϕ is its normalising constant. Since Δ depends on θ only through $\log q_\theta$, the gradient is

$$\nabla_\theta \mathcal{L}_{\text{RTB}} = 2 \mathbb{E} [(\log Z_\phi + \Delta(\tau)) \nabla_\theta \log q_\theta(\tau)], \quad (8)$$

a scalar residual multiplying a score-function term. Minimising over ϕ alone gives the closed form $\log Z^* = -\mathbb{E}[\Delta]$. If $\log Z_\phi$ lags this optimum by ϵ , every residual carries the same offset and (8) contributes $2\epsilon \mathbb{E}[\nabla_\theta \log q_\theta]$, which displaces all log-probabilities in a common direction without distinguishing between trajectories. This motivates the separate, larger learning rate we use for $\log Z_\phi$ (Appendix C).

273 A.3 Cancellation of the coordinate term

274 Substituting (6) into Δ , the log-ratio separates:

$$\log \frac{q_\theta(\tau)}{p_{\text{prior}}(\tau)} = \underbrace{\sum_{t=1}^T \log \frac{q_{\text{atom}}(a_t | s_{t-1})}{p_{\text{atom}}(a_t | s_{t-1})}}_{\text{accumulated during rollout}} + \underbrace{\sum_{t=1}^T \log \frac{p_{\text{diff}}(x_t | z_t^\theta)}{p_{\text{diff}}(x_t | z_t^0)}}_{=: B(\tau)}. \quad (9)$$

275 The coordinate model p_{diff} is frozen in all our experiments, so the two terms of $B(\tau)$ differ only
 276 through their conditioners. Under the head-only scope (PROJ), enc_1 and enc_2 are also frozen, so z_t^θ
 277 and z_t^0 are the same function evaluated on the same argument ($a_{\leq t}, x_{< t}$). Every term of $B(\tau)$ is
 278 therefore $\log 1 = 0$ identically, and the atom-only log-ratio is exact rather than approximate. This is
 279 not an assumption about magnitudes; it follows from enc_1 and enc_2 being untouched. It also permits
 280 a computational saving: a single forward pass through enc_1 serves both policy and prior.

281 Under the ATOM and FULL scopes the trunk is trainable, $z_t^\theta \neq z_t^0$, and $B(\tau)$ no longer vanishes.
 282 Taking $\log p_{\text{diff}}$ to be locally Lipschitz in its conditioner with constant L_t on the relevant region,

$$|B(\tau)| \leq \sum_{t=1}^T L_t \|z_t^\theta - z_t^0\|. \quad (10)$$

283 We do not estimate L_t , which would require differentiating through the 18-step sampler. We instead
 284 $\log \|h_t^\theta - h_t^0\|/\|h_t^0\|$ at every step as a relative proxy for the displacement in (10), since h_t is the
 285 input to both W_{proj} and enc_2 . This quantity is 0 by construction under PROJ; the values observed
 286 under ATOM and FULL are reported in Appendix D. It bounds neither $B(\tau)$ nor the resulting bias, and
 287 we report it only so that the size of the violation is measured rather than assumed. Results obtained
 288 under those two scopes should be read as approximate-ratio results.

289 A.4 LoRA on the projection is a rank- r linear guide

290 A LoRA adapter of rank r on a weight $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ replaces it with $W' = W + \frac{\alpha}{r} BA$, where
 291 $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$, with B initialised to zero so that $W' = W$ at the start of training.
 292 Applied to W_{proj} , the induced change in the atom-type logits is

$$\delta \ell(h) = \frac{\alpha}{r} BA h, \quad (11)$$

293 which is a linear map of h into logit space of rank at most r . Comparing with Equation (3), LoRA
 294 is therefore a guide g_θ constrained to be linear and rank-limited, but unconstrained in norm. The
 295 residual guide is the opposite: nonlinear and full-rank, but with a norm that training keeps small
 296 relative to $\|\ell\|$. Sweeping r traverses a capacity axis that the residual guide does not, which is the
 297 reason the two families appear together in Table 1.

298 A.5 Properties of the AUC metric

299 Let $f_k(n)$ be the mean of the k highest scores among the first n oracle calls, and $\text{AUC}_k(B) =$
 300 $B^{-1} \int_0^B f_k(n) dn$. Since the set of scored molecules grows with n , f_k is non-decreasing, and
 301 therefore

$$\text{AUC}_k(B) \leq f_k(B), \quad (12)$$

302 with equality only if the optimum is found at the first call. The difference between the two measures
 303 how quickly a method reaches its best molecules, which is why we report both. In practice we
 304 evaluate f_k on a grid of spacing 100 anchored at $f_k(0) = 0$ and integrate by the trapezoid rule.

305 A consequence relevant to Figure 1 is that the two quantities cannot be compared directly. A
 306 configuration reported by $f_k(B)$ will appear at least as high as one reported by $\text{AUC}_k(B)$ with an
 307 identical score distribution. For a non-adaptive sampler — a frozen policy or a static dataset, neither
 308 of which improves within a budget — no ordering of the sample is privileged, so AUC_k is well
 309 defined in expectation over random permutations and can be estimated by resampling. We report
 310 $f_k(B)$ for those cases in the main text and give the permutation estimates in Appendix D.

311 A.6 Unbiased subsampling of the trajectory log-probability

312 Retaining the autograd graph for every step of a long rollout is memory-bound. Let $u_t = \log q_\theta(a_t |$
 313 $s_{t-1})$ and $S = \sum_t u_t$. With $m_t \sim \text{Bernoulli}(\rho)$ drawn independently of the trajectory, we form

$$\tilde{S} = \sum_t \left[\text{sg}(u_t) + (u_t - \text{sg}(u_t)) \frac{m_t}{\rho} \right], \quad (13)$$

314 where sg denotes the stop-gradient operator. Two properties hold. The value is exact, $\tilde{S} =$
 315 $\sum_t \text{sg}(u_t) = S$, so the scalar residual in (8) is not an estimate. The gradient is $\nabla \tilde{S} = \rho^{-1} \sum_t m_t \nabla u_t$,
 316 and since m is independent of the trajectory, $\mathbb{E}[\nabla \tilde{S} | \tau] = \nabla S$. The estimator is therefore unbiased
 317 conditional on τ , at the cost of variance inflation of order $(1 - \rho)/\rho$ per step, in exchange for retaining
 318 approximately ρT of the graph. We use $\rho = 0.25$ for the FULL scope and $\rho = 1$ elsewhere.

319 B Architectures and Implementation

320 B.1 The frozen prior

321 Quetzal is a decoder-only transformer with 12 layers split into two stacks of 6, hidden width 768, 12
 322 attention heads, and learned positional embeddings over a block size of 512. The first stack, enc_1 ,
 323 embeds atom types, coordinates and a Fourier featurisation of the coordinates, and produces the
 324 hidden state h_t from which $W_{\text{proj}} \in \mathbb{R}^{128 \times 768}$ (no bias) emits atom-type logits. The second stack,
 325 enc_2 , re-embeds the sampled atom type and produces the conditioner z_t for the coordinate model, an
 326 adaptive-layernorm MLP of width 1536 and depth 6 trained as an EDM-style denoiser and sampled
 327 with 18 Heun steps. Atom types are atomic numbers over a vocabulary of 128, with three reserved
 328 tokens (STOP = 0, PAD = 126, GEN = 127); a sequence terminates at the first STOP. We use the
 329 GEOM-Drugs checkpoint throughout and take the exponential-moving-average weights, which are
 330 the weights the published sampling results use.

331 B.2 Guide architectures

332 All three guides are functions of h_t alone and are applied at every atom decision. Each is zero-
 333 initialised in its final layer so that the guided logits equal the prior’s logits exactly at initialisation,
 334 which avoids any regression relative to the frozen model at the start of training.

335 **Residual guide.** An MLP $g_\theta : \mathbb{R}^{768} \rightarrow \mathbb{R}^{128}$ with one hidden layer of width 512 and SiLU
 336 activation, added directly to the logits as in Equation (3). The final linear layer has zero weight and
 337 bias.

338 **Temperature and gain guide.** Two scalar heads are added to the residual guide, each an MLP of
 339 hidden width 128:

$$T_\theta(h) = 1 + (\text{softplus}(w_T^\top h + b_T) - \log 2), \quad \gamma_\theta(h) = \text{softplus}(w_\gamma^\top h + b_\gamma). \quad (14)$$

340 The offset $-\log 2$ makes $T = 1$ at zero pre-activation, and b_γ is initialised to $\text{softplus}^{-1}(1)$ so that
 341 $\gamma = 1$. The intended behaviour is $T > 1$ on decisions where the prior is over-confident, softening it
 342 enough for the residual to compete. The implementation applies $\text{clamp}(T, \min=1)$ before dividing,
 343 which prevents sharpening but also prevents the head from expressing any learned value below 1; the
 344 consequences are discussed in Appendix F.

345 **Hidden guide.** An MLP $\delta_\theta : \mathbb{R}^{768} \rightarrow \mathbb{R}^{768}$ of the same shape, added to the hidden state before the
 346 frozen projection, as in Equation (5). The motivation is that W_{proj} amplifies displacements in the
 347 directions it uses: empirically, a δ of norm 0.1 produces a change of ≈ 46 in logit space, against
 348 ≈ 1.3 for an output residual of the same norm. The guide holds a reference to W_{proj} and returns
 349 guided logits directly; for compatibility with call sites expecting an additive residual it can also return
 350 $W_{\text{proj}}(h + \delta_\theta(h)) - W_{\text{proj}}h$.

Table 2: Trainable parameters by scope. $d = 768$, $|V| = 128$. LoRA counts are $r(d + |V|)$.

Scope	Modules updated	Parameters
PROJ	W_{proj}	98,304
PROJ + LoRA-4	rank-4 adapter on W_{proj}	3,584
PROJ + LoRA-16	rank-16 adapter	14,336
PROJ + LoRA-64	rank-64 adapter	57,344
ATOM	W_{proj} , enc_1 , embeddings	≈ 43 M
FULL	all of the above, enc_2 , coordinate denoiser	≈ 85 M

B.3 Fine-tuning scopes

Only PROJ preserves the exact cancellation of Appendix A.3; the other two are approximate-ratio settings. Two copies of the model are held in memory during fine-tuning, one frozen for the prior term of the ratio and one trainable. Head and trunk parameters are placed in separate optimiser groups, with the trunk at one tenth of the head’s learning rate, and $\log Z_\phi$ in a third group at a substantially larger rate for the reason given in Appendix A.2. Checkpoints store only trainable parameters, with frozen weights restored from the base checkpoint on load.

B.4 Sampling and reward evaluation

Generation is autoregressive without a key-value cache, so the cost of a rollout is quadratic in molecule length; for GEOM-sized molecules the dominant per-step cost is nonetheless the reward evaluation, which runs on CPU and requires bond perception from 3D coordinates followed by the benchmark’s own scoring function. Rewards are computed at $\beta = 1$ and multiplied by β inside the trainer. Molecules that fail bond perception or scoring return a fixed floor value of -5 in log space; the fraction of samples above this floor is reported alongside every result, since a flat mean log-reward is otherwise ambiguous between a policy that is not steering and one whose samples are largely invalid. RDKit is pinned at version 2023.03.3 throughout, as later versions alter bond perception and therefore every validity and stability figure we report.

C Hyperparameters and Compute

C.1 Compute

All experiments were run on a single NVIDIA A100 SXM with 80 GB of memory. Fine-tuning holds two copies of the model in memory, one frozen to supply the prior term of the trajectory ratio and one trainable; together these occupy under 1 GB in bfloat16, and the binding memory cost is instead the autograd graph retained over the rollout, which grows with batch size and molecule length. Under the FULL scope we retain a random $\rho = 0.25$ fraction of the per-step graph (Appendix A.6), and reduce the batch size to 8.

Table 3: Compute per configuration. [FILL] entries from recorded wall-clock.

Configuration	Batch	Steps	Molecules	GPU-hours
Guide sweep (per run)	128	[FILL]	[FILL]	[FILL]
FT PROJ	64	600	38,400	[FILL]
FT ATOM	64	600	38,400	[FILL]
FT FULL	8	600	4,800	[FILL]
FT LoRA (all ranks)	64	600	38,400	[FILL]
Dumping (per checkpoint)	—	—	5,000	[FILL]

C.2 Guide training

C.3 Fine-tuning

Reward is computed at $\beta = 1$ and multiplied by the current β inside the trainer. The learning rate for $\log Z_\phi$ is three orders of magnitude larger than the weight learning rate; the reason is given in

Table 4: Guide-training hyperparameters. Values not listed are the defaults of the configuration dataclass.

Setting	Value
Guide hidden width / depth	512 / 2 layers, SiLU
Final-layer initialisation	zero (weight and bias)
Optimiser	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.99$
Learning rate (guide)	[FILL]
Learning rate (log Z / flow head)	[FILL]
Weight decay	[FILL]
Gradient clipping	1.0
EMA decay (guide)	[FILL]
Batch size	128 trajectories
Steps per epoch / epochs	100 / 8
Reward exponent β	{1, 10} (Osimertinib additionally 100)
Sampling temperature / ε	[FILL]
Maximum molecule length	192 atoms
Diffusion steps	18
Invalid log-reward floor	-5
Replay fraction / capacity	0.25 / 10,000, reward-prioritised
Seeds	0, 42, 100

Table 5: Fine-tuning hyperparameters. The trunk learning rate applies only to the ATOM and FULL scopes.

Setting	Value
Objective	RTB
Learning rate (head)	1×10^{-5} (FULL: 5×10^{-6})
Learning rate (trunk)	$0.1 \times$ head
Learning rate (log Z_ϕ)	1×10^{-2}
Optimiser	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.99$, $\varepsilon = 10^{-12}$
Weight decay	0
Gradient clipping	1.0
LR warmup	100 steps, linear
Batch size	64 (FULL: 8)
Steps	600
β schedule	linear $1 \rightarrow 10$, reaching 10 at step 400
LoRA α / dropout	16 / 0
LoRA target	W_{proj}
Log-probability subsampling ρ	1.0 (FULL: 0.25)
Sampling temperature / ε	1.0 / 0
Precision	bf16 mixed
Seeds	[FILL: SINGLE SEED AT PRESENT]

Appendix A.2. Sampling during training is on-policy at temperature 1 with no ε -uniform mixing, so that the recorded molecule stream is a clean sample of the policy and can be scored as a goal-directed run without further correction.

C.4 Software

RDKit is pinned at 2023.03.3; later versions alter bond perception and therefore every validity, stability and conversion-rate figure reported here. GuacaMol scoring functions are used unmodified, and the same scoring object is used at training and evaluation time so that recorded and re-scored rewards agree (Appendix ??).

D Complete Results

D.1 Guide sweep

Table 6: Guide sweep on the MPO benchmarks, mean $\pm$ standard deviation over three seeds (0, 42, 100). $\Delta\text{top-10}$ is guided minus frozen prior on the benchmark’s own scale.

Reward	Guide	Obj.	β	$\Delta\text{top-10}$	$\text{FCD}_{g b}$	Uniq.	Parse
Osimertinib	base	DB	10	[FILL]	[FILL]	[FILL]	[FILL]
Osimertinib	hidden	DB	10	[FILL]	[FILL]	[FILL]	[FILL]
Osimertinib	tempgain	DB	10	[FILL]	[FILL]	[FILL]	[FILL]
Osimertinib	hidden	RTB	10	[FILL]	[FILL]	[FILL]	[FILL]
Perindopril	base	DB	10	[FILL]	[FILL]	[FILL]	[FILL]
Perindopril	hidden	DB	10	[FILL]	[FILL]	[FILL]	[FILL]

[FILL: REMAINING ROWS FROM MASTER_LONG.CSV]

D.2 Flip diagnostics

Table 7: Coupled-flip diagnostics at temperature 1, averaged over all runs of each guide type across rewards, objectives and β . High-margin decisions are those with a top-1 logit margin above 8.

Guide	Runs	Delivery	Flip (all)	Flip (high)	Flip (low)	Margin
base	22	1.000	0.0079	0.0002	0.0276	16.39
hidden	30	1.000	0.0098	0.0006	0.0330	16.38
tempgain	16	1.000	0.0049	0.0000	0.0174	16.36

Across all three architectures, 71.7–71.9% of decisions fall in the high-margin regime. Delivery is exactly 1.000 throughout, confirming that the guided logits differ from the prior’s at every state and that the near-zero high-margin flip rate is not a wiring failure.

D.3 Fine-tuning

[†] Compute limits required a batch size of 8 for the FULL scope against 64 elsewhere, so at the same step count it reached 4,800 oracle calls rather than 38,400. Since top- k grows with the number of calls, this run is not comparable on a capacity axis and is excluded from Figure 3. [FILL: REPLACE WITH A MATCHED-BUDGET RE-RUN ONCE AVAILABLE.]

D.4 Novelty and diversity

D.5 Trunk drift under approximate-ratio scopes

[FILL: REPORT DIAG/ZPREFIX_DRIFT AT THE END OF TRAINING FOR THE ATOM AND FULL RUNS, AS THE RELATIVE DISPLACEMENT DEFINED IN APPENDIX A.3. THIS IS LOGGED DURING TRAINING AND REQUIRES NO ADDITIONAL EXPERIMENT, ONLY EXTRACTION FROM THE RUN LOGS.]

E The Nitrogen Positive Control

E.1 Motivation

A null result on a benchmark objective is ambiguous between two explanations: the training loop does not work, or the objective is not reachable by the intervention. To separate them we constructed a reward with the properties a logit-level guide should find easiest. The nitrogen reward is the fraction of heavy atoms that are nitrogen,

$$R_N(x) = \frac{|\{i : Z_i = 7\}|}{|\{i : Z_i > 1\}|}, \quad (15)$$

Table 8: Fine-tuning at a matched budget of 10,000 oracle calls, top-10 mean. Trainable-parameter counts follow Table 2. The FULL run reached only 4,800 calls and is reported separately.

Configuration	Params	Calls	Top-10	Uniq.	Quality
<i>Osimertinib MPO</i> (GEOM best-of-10k: 0.7912)					
LoRA-4 PROJ	3,584	10,000	0.8080	0.997	0.88
LoRA-16 PROJ	14,336	10,000	0.7913	0.998	0.94
LoRA-64 PROJ	57,344	10,000	0.8101	0.993	0.91
LoRA-64 PROJ (alt. init)	57,344	10,000	0.8042	0.997	0.79
PROJ ($\beta = 10$)	98,304	10,000	0.8012	0.997	0.89
PROJ ($\beta = 20$)	98,304	10,000	0.8173	0.854	0.96
LoRA-4 ATOM	43.0 M	9,664	0.8021	0.963	0.79
LoRA-16 ATOM	43.0 M	10,000	0.8029	0.999	0.86
LoRA-64 ATOM	43.1 M	10,000	0.8044	0.984	0.89
ATOM	43.1 M	10,000	0.7964	0.998	0.84
FULL [†]	85 M	4,800	0.7835	0.999	0.90
<i>Perindopril MPO</i> (GEOM best-of-10k: 0.4761)					
LoRA-4 PROJ	3,584	10,000	0.4641	0.997	0.95
LoRA-16 PROJ	14,336	10,000	0.4789	0.998	0.96
LoRA-64 PROJ	57,344	10,000	0.4596	0.998	0.96
PROJ	98,304	10,000	0.4722	0.997	0.92
LoRA-4 ATOM	43.0 M	9,600	0.4604	0.998	0.93
LoRA-16 ATOM	43.0 M	9,600	0.4633	0.998	0.91
LoRA-64 ATOM	43.1 M	9,600	0.4720	0.998	0.94
ATOM	43.1 M	9,600	0.4678	0.998	0.84
FULL [†]	85 M	4,800	0.4492	0.998	0.89

Table 9: Distribution statistics of the top-100 molecules, fine-tuned configurations. Ranges are across configurations within each benchmark.

Statistic	Osimertinib	Perindopril
3D-to-SMILES conversion	0.937–0.951	0.928–0.944
Uniqueness among valid	0.854–0.999	0.997–0.998
Internal diversity	0.734–0.865	0.790–0.825
Murcko scaffold diversity	0.87–1.00	0.96–1.00
Compound-quality pass rate	0.79–0.96	0.84–0.96
Novelty vs. GEOM-Drugs	0.91–0.97	0.86–0.96
Nearest-neighbour Tanimoto	0.401–0.432	0.426–0.473
GEOM-Drugs reference quality	0.97	0.96

which is dense (over 90% of samples score above the invalid floor, against 2–6% for the assembled MPO objectives), monotone in a single atom-level decision, and requires no geometric or topological structure. It is decomposable in exactly the way an atom-type guide can exploit: every atom placed contributes independently to the score. It is not a meaningful design objective and we do not report it as a result about molecular optimisation; its role is to establish that the machinery steers when the objective is reachable.

E.2 Results

The best configuration, a hidden guide trained with detailed balance at $\beta = 10$, raises the top-10 nitrogen fraction from 0.452 to 0.983, an increase of $+0.791 \pm 0.099$ across seeds. This is two to three orders of magnitude larger than the largest effect observed on either MPO benchmark, where $\Delta_{\text{top-10}}$ does not exceed 0.005 for any guide. The training loop, the guide architectures and the reward plumbing therefore steer the sampler when the objective admits it, and the null results in Section 5 cannot be attributed to an implementation failure.

Table 10: Nitrogen control, mean $\pm$ standard deviation over three seeds. The frozen prior scores 0.4519 (top-10) and 0.1189 (mean). Δ is guided minus prior.

Guide	Obj.	β	Replay	$\Delta_{\text{top-10}}$	Top-10	Mean	FCD _{g b}
hidden	DB	10	off	$+0.791 \pm 0.099$	0.983	0.152	1.170
hidden	DB	1	off	$+0.704 \pm 0.179$	0.918	0.129	0.496
hidden	DB	10	on	$+0.490 \pm 0.103$	0.751	0.137	0.604
hidden	DB	1	on	$+0.353 \pm 0.159$	0.663	0.128	0.356
base	DB	10	on	$+0.372 \pm 0.111$	0.673	0.146	0.896
base	DB	10	off	$+0.296 \pm 0.048$	0.619	0.145	0.854
base	DB	1	on	$+0.065 \pm 0.019$	0.484	0.126	0.224
base	DB	1	off	$+0.041 \pm 0.011$	0.473	0.123	0.186
hidden	revKL	10	off	$+0.077 \pm 0.095$	0.489	0.132	0.322
hidden	RTB	10	off	$+0.061 \pm 0.149$	0.481	0.132	0.482
hidden	revKL	1	off	-0.004 ± 0.080	0.446	0.122	0.076
hidden	RTB	1	off	-0.003 ± 0.080	0.447	0.122	0.102
tempgain	DB	1	off	-0.004 ± 0.080	0.446	0.122	0.086

E.3 Three observations

The objective matters here, unlike on the MPO benchmarks. Detailed balance steers strongly while relative trajectory balance and reverse KL do not ($+0.791$ against $+0.061$ and $+0.077$ at matched β and architecture). This is the one place in our results where the choice of loss changes the outcome, and it is worth stating alongside the observation in Section 5 that on the MPO benchmarks the losses agree. A plausible reading is that DB’s per-state flow head can propagate a dense, decomposable reward backwards along the trajectory, whereas RTB assigns credit only through a single terminal residual and a global normalising constant; when the reward decomposes over atoms, the former has structure to exploit that the latter discards. [FILL: THIS IS AN INTERPRETATION, NOT A MEASUREMENT. IT COULD BE TESTED BY EVALUATING DB AND RTB ON A REWARD OF MATCHED DENSITY BUT NON-DECOMPOSABLE STRUCTURE.]

Steering does not require leaving the valid region. Parse rate holds at 0.93–0.94 across every nitrogen configuration, matching the frozen prior’s 0.929, and uniqueness stays above 0.98. The distributional shift is nonetheless large, with FCD against the prior reaching 1.17 for the best configuration against 0.29 averaged over the MPO runs. A dense reward is therefore reachable without the validity–reward trade-off described in Section 5, which supports reading that trade-off as a property of how far the objective sits from the prior rather than an inevitable cost of steering.

Replay reduces steering here. For the hidden guide at $\beta = 10$, replay lowers $\Delta_{\text{top-10}}$ from $+0.791$ to $+0.490$. Replay is designed to improve mode discovery under sparse rewards; on a dense reward it instead concentrates training on previously-sampled high-reward trajectories and appears to slow the policy’s movement. This is consistent with our observation on the MPO benchmarks that replay changes nothing, since there the bound is not set by exploration.

E.4 The contrast with the MPO benchmarks

Table 11: Best $\Delta_{\text{top-10}}$ by reward, over all guide architectures, objectives and β . Reward density is the fraction of generated molecules scoring above the invalid floor.

Reward	Density	Best $\Delta_{\text{top-10}}$	Structure
Nitrogen fraction	> 0.90	$+0.791 \pm 0.099$	dense, atom-decomposable
Osimertinib MPO	0.02–0.06	[FILL]	sparse, multi-property
Perindopril MPO	[FILL]	[FILL]	sparse, multi-property

The same code, architectures and objectives produce a change of $+0.79$ on one reward and under 0.01 on the others. The difference between them is not the method but the reward: one is dense and decomposes over the decisions the guide controls, the others are sparse and depend on structure the guide can only influence indirectly through the first few atom choices. We take this as evidence that

the bound reported in Section 5 is a property of the interaction between the prior and the objective, rather than of the steering machinery.

F The Temperature and Gain Guide

F.1 Motivation

The residual guide of Equation (3) competes against logits whose top-1 margin averages 16.4. The temperature and gain architecture was designed to change the terms of that competition rather than the size of the residual. Dividing the prior’s logits by a learned per-state temperature $T_\theta(h) > 1$ compresses the margin, so that a residual of unchanged magnitude can contest decisions it could not otherwise reach; the gain $\gamma_\theta(h)$ then scales the residual where the guide is confident. Both are learned per state rather than set globally, because a uniform softening applies where it helps and where it destroys validity alike. An earlier scale sweep had shown that multiplying the residual by a constant factor raises the effect size up to roughly $4\times$ and then reverses as validity collapses, which motivated making the strength state-dependent.

The intended behaviour is therefore $T > 1$ selectively: large on decisions where the prior is over-confident but the alternatives are chemically reasonable, and close to 1 elsewhere.

F.2 The implementation defect

The forward pass computes

$$\tilde{\ell}(h) = \frac{W_{\text{proj}}h}{\text{clamp}(T_\theta(h), \min = 1)} + \gamma_\theta(h)g_\theta(h). \quad (16)$$

The clamp was included to prevent the head from sharpening the prior, which was judged an unhelpful direction. It has the side effect that no learned value below 1 has any effect: the head’s output is replaced by the constant 1, the division becomes the identity, and the gradient with respect to the temperature parameters is zero over that region.

Reading $T_\theta(h)$ directly from trained checkpoints shows that this is the region the head occupies. Across every margin bin and all three component guides examined, the learned temperature lies between 0.73 and 0.80, is flat with respect to the prior’s margin, and never approaches 1 from above (Figure 8). The head learned to sharpen the prior — the opposite of the intended direction — and was clamped to the identity at every state. The temperature mechanism was therefore inactive throughout training and evaluation, and the runs labelled TEMPGAIN report a residual guide with a learned gain.

F.3 Consequences for the reported results

The flip diagnostics are consistent with an inactive mechanism. Averaged over 16 runs at temperature 1, the TEMPGAIN guide moves less probability mass than the plain residual guide it wraps, not more: mean total variation 0.0040 against 0.0068, mean KL 0.0021 against 0.0063, and sampled-flip rate 0.0050 against 0.0079. Its high-margin flip rate is 0.00004, the lowest of the three architectures. On the nitrogen control it is the only architecture with no measurable effect ($\Delta_{\text{top-10}} = -0.004 \pm 0.080$, against $+0.79$ for the hidden guide under the same objective).

We report TEMPGAIN results throughout for completeness, but they should be read as a gain-scaled residual guide and not as a test of prior softening. Two consequences follow. First, no conclusion in this work rests on the temperature mechanism: the bound is established by the residual and hidden guides and by fine-tuning, all of which are unaffected. Second, prior softening remains untested. Whether a genuinely bidirectional temperature would relax the bound, or would trade margin for validity in the manner the scale sweep suggests, is open.

F.4 What a corrected experiment requires

[FILL: THIS EXPERIMENT HAS NOT BEEN RUN.] Removing the clamp is a one-line change, but a corrected run needs more than that. The parameterisation should permit T on both sides of 1 with the identity as its initialisation, for instance $T_\theta(h) = \exp(u_\theta(h))$ with u zero-initialised, so that neither direction is structurally favoured. Validity should be logged per state alongside T , since

496 the mechanism’s premise is that softening is safe where the alternatives are reasonable and unsafe
 497 elsewhere, and that premise is itself untested. The natural comparison is against the hidden guide
 498 at matched β and objective on both the nitrogen control and the MPO benchmarks: the control
 499 establishes whether the corrected mechanism steers at all, and the benchmarks whether softening
 500 reaches decisions the other architectures cannot.

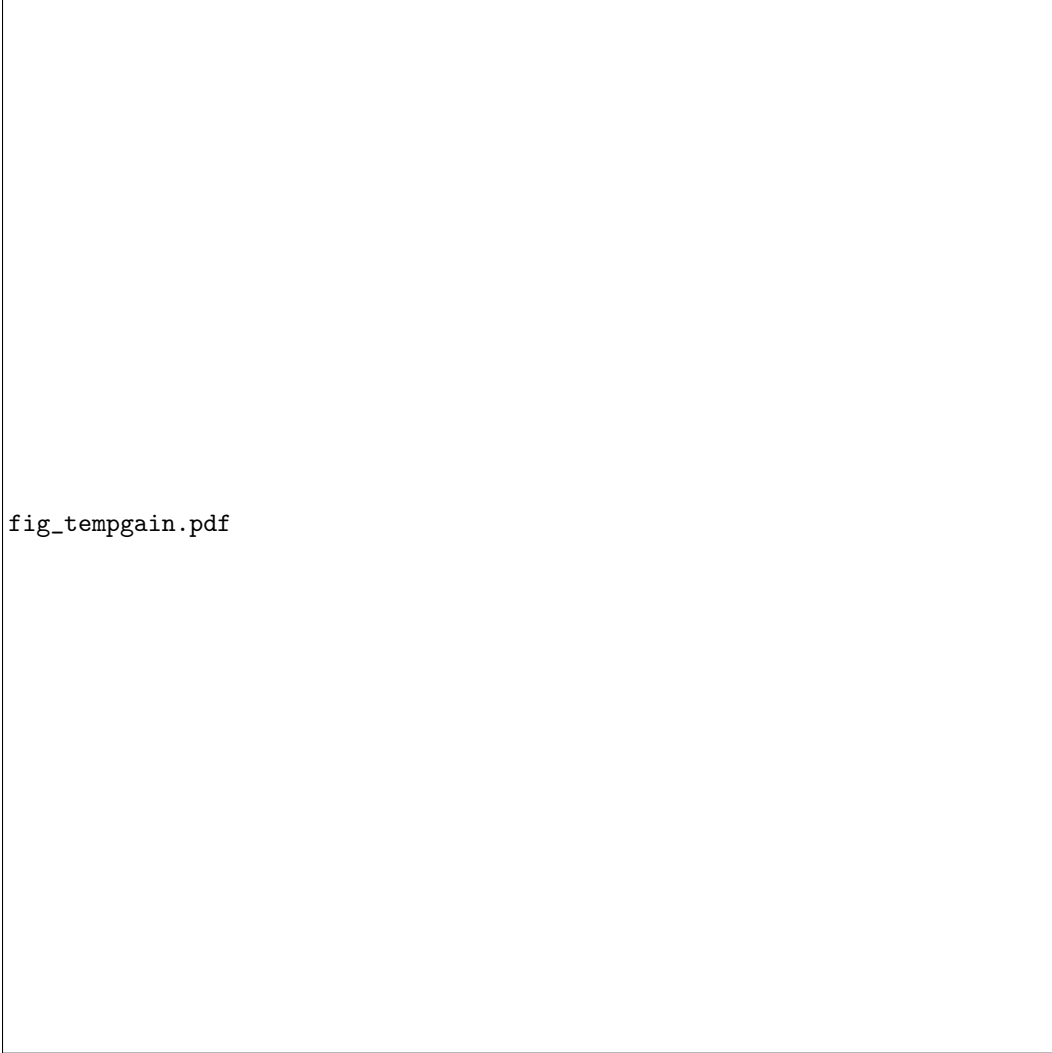


Figure 4: **The learned temperature never exceeds one.** Left: $T_\theta(h)$ read from trained checkpoints, binned by the prior’s top-1 logit margin, for three component guides. The dashed line marks $T = 1$, above which the mechanism would soften the prior; the learned values sit between 0.73 and 0.80 everywhere and are flat in the margin. Since the forward pass applies $\text{clamp}(T, \min=1)$, the effective temperature was 1 at every state. Right: sampled-flip rate by margin for the same checkpoints, showing the decay to zero above a margin of 8 that all architectures share.

501 G Excluded Runs and Corrected Defects

502 This appendix records runs excluded from the reported results and defects found and fixed during the
 503 work. Several were caught only because a diagnostic disagreed with an aggregate metric, which is
 504 part of why the flip diagnostics of Section 4 exist.

505 **G.1 Excluded runs**

506 **Composition experiments (flow route).** A set of composed-guide runs sampled through a flow-
507 based routing path in which the guide residual was computed but not applied to the logits. The
508 rollout diagnostics show $\|g_\theta(h)\| = 0.000$ exactly, at every state and for all four components, and
509 the corresponding flip rate is identically zero at every sequence position. This is the delivery-failure
510 signature rather than a bound: the equivalent runs on the policy route, with the same checkpoints,
511 show residual norms of 0.37 to 8.18 and non-zero flip rates. The flow-route runs are excluded from
512 all reported results.

513 **Atom-stability objective.** Runs targeting EDM atom stability produced $\Delta_{\text{top-10}}$ of exactly 0.000
514 for both the residual and hidden guides. Molecular stability, the fraction of molecules in which
515 every atom satisfies its valence, is below 0.05 throughout, which is expected rather than anomalous:
516 for molecules of 80–100 heavy atoms, even a per-atom stability of 0.95 gives $0.95^{80} \approx 0.017$.
517 The objective is close to saturated under the prior and offers little gradient, and we exclude it as
518 uninformative rather than as evidence.

519 **Undertrained sanity run.** The nitrogen control described in Appendix E includes a fine-tuning run
520 of two epochs, used only to check that the fine-tuning loop trains at all before committing compute to
521 the benchmark runs. It is excluded from all comparisons.

522 **Under-budget full fine-tuning.** Compute limits required a batch size of 8 for the FULL scope
523 against 64 elsewhere, so at the same step count it recorded 4,800 oracle calls rather than 38,400. Since
524 top- k increases with the number of calls, this run is not comparable on a capacity axis and is excluded
525 from Figure 3, though it is reported in Table 8. [FILL: REPLACE WITH A MATCHED-BUDGET
526 RE-RUN.]

527 **H Additional Figures**

528 Figures in this appendix support claims made in the main text but are not needed to follow the
529 argument. Each caption states which section it supports.

530 **H.1 The margin ceiling**

531 **H.2 Residual scale and the validity limit**

532 **H.3 The temperature and gain guide**

533 **H.4 Composition**

534 **H.5 Distribution health during fine-tuning**

535 **H.6 Sample efficiency**

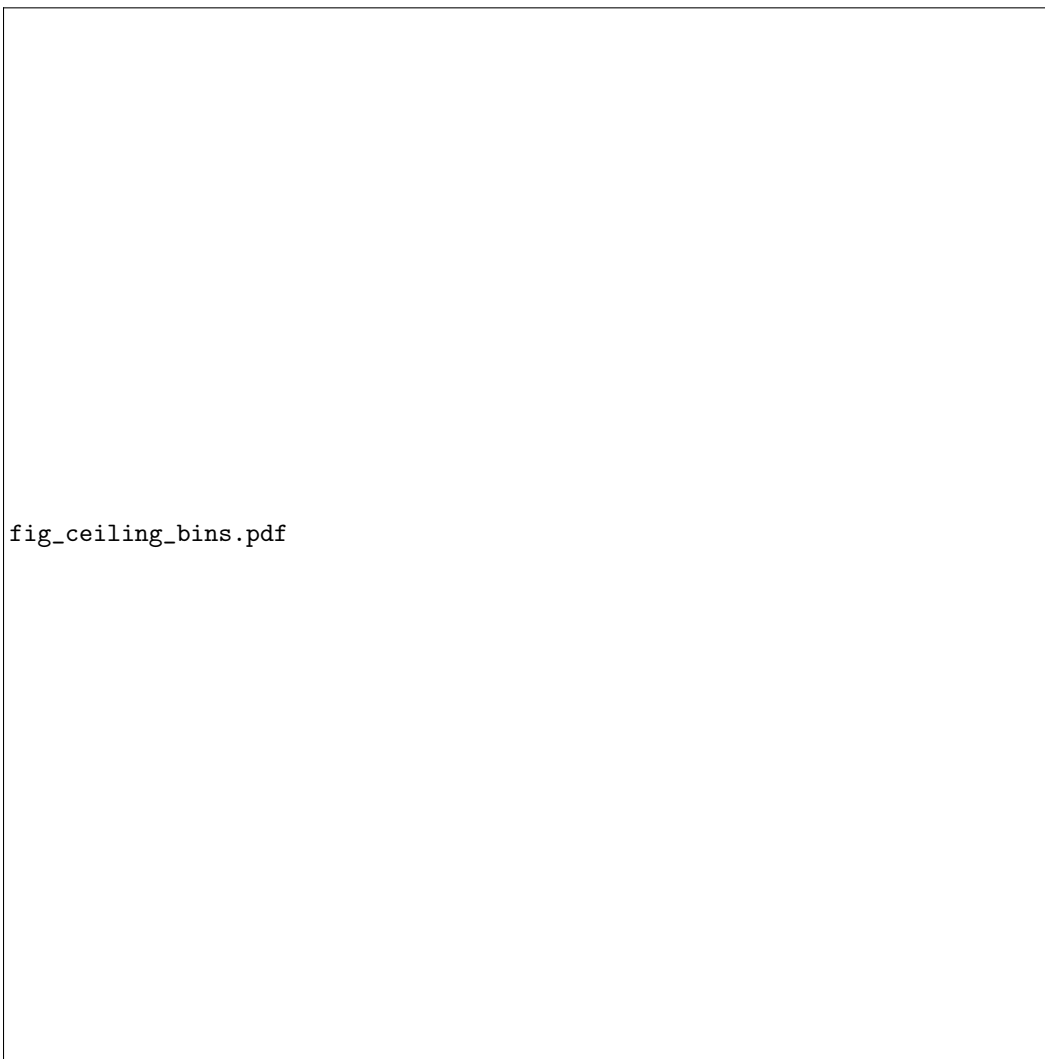


Figure 5: **Flip rate against the prior’s top-1 logit margin.** Supports Section 5. Left: coupled sampled-flip rate for each guide, binned by the prior’s margin at the decision, with the grey bars giving the fraction of decisions falling in each bin (right-hand axis). The flip rate falls to zero above a margin of roughly 4, while the majority of decisions lie above 8. Right: standard deviation of the log-reward over prior samples for each leaf scorer of the Osimertinib objective, with the dashed line marking the threshold below which an axis carries no gradient. One component has zero variance over reachable molecules and is flat by construction; a guide trained against it cannot steer, which is why the benchmark runs use the assembled objective rather than its components (Appendix G).

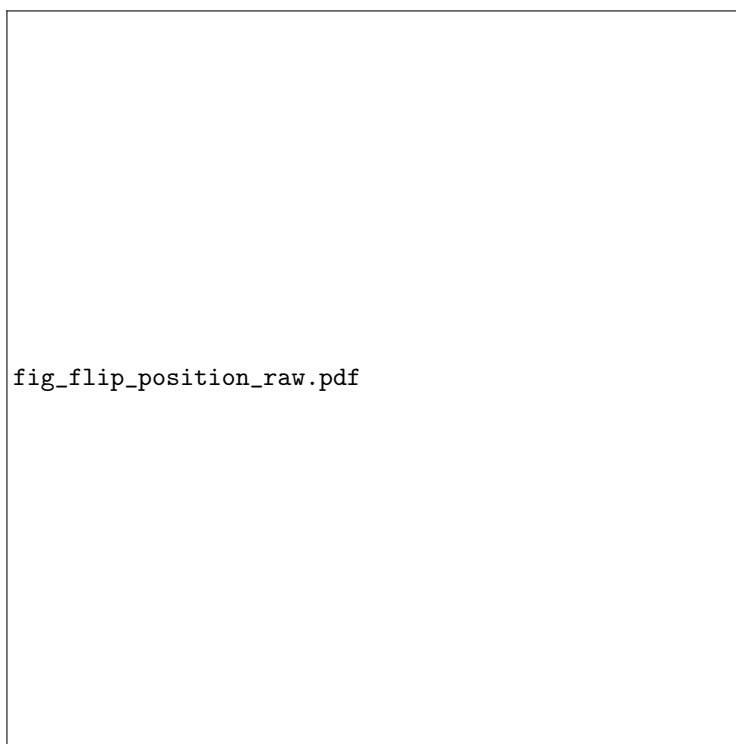


Figure 6: **Sampled-flip rate by sequence position, individual guides.** Supports Figure 2, which averages over configurations. Each line is one guide evaluated on trajectories rolled by the frozen prior. The decay from the first atom is present in every guide, though its magnitude at position zero varies by roughly a factor of six across them; all converge to near-zero by position four. Position indices run over the first eight atom decisions only, the limit of the diagnostic as originally recorded.



Figure 7: **Effect size against residual scaling.** Supports Section 5. Left: Wasserstein-1 distance between the guided and prior reward distributions, and the mean log-reward shift, as the trained guide residual is multiplied by a constant factor at sampling time. Both rise to a maximum near $4\times$ and then fall, with the mean shift turning negative: scaling the residual past that point moves the sampler off the region where its outputs remain valid faster than it gains reward. Right: residual norm relative to the prior logit norm, and $\text{KL}(\text{guided} \parallel \text{prior})$, for each guide. The residual is a small fraction of the prior’s logit magnitude in every case.

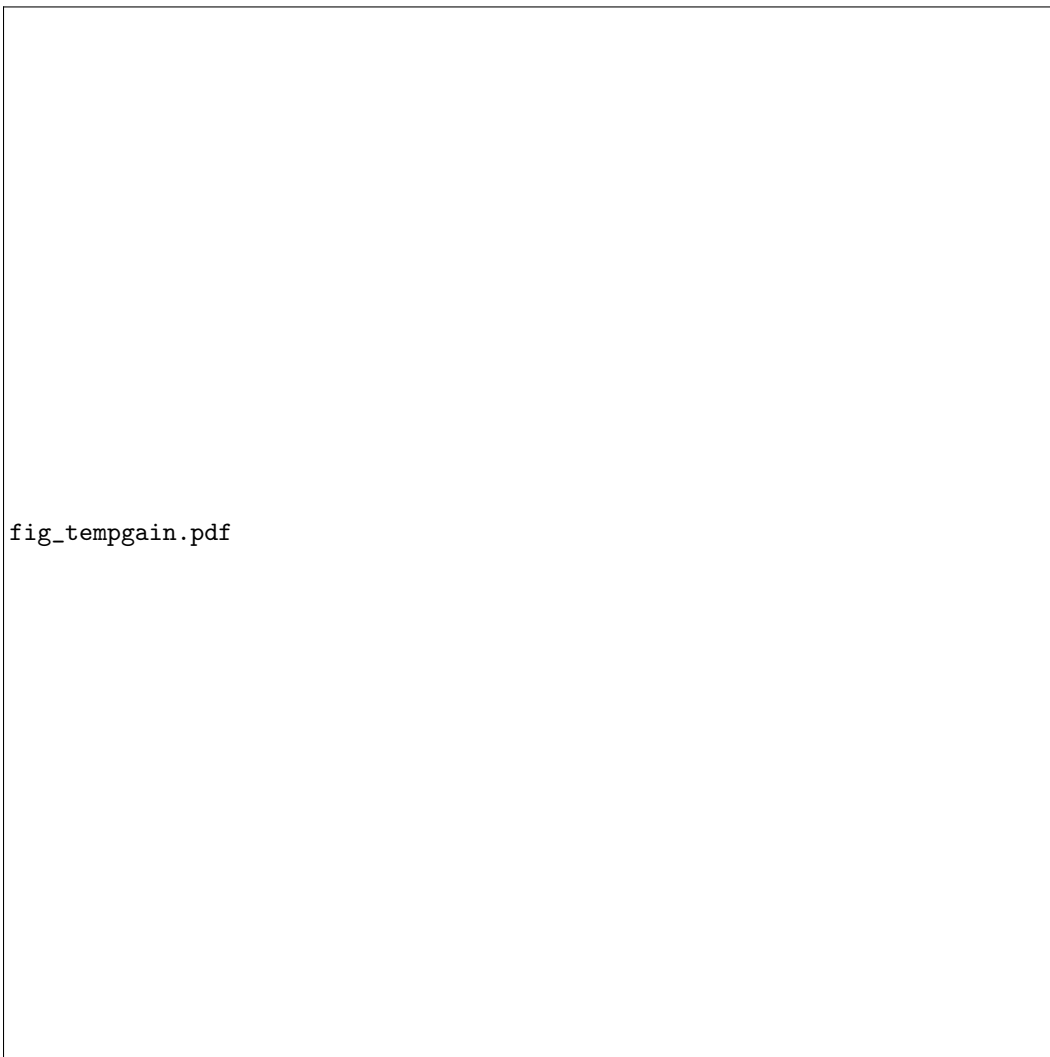
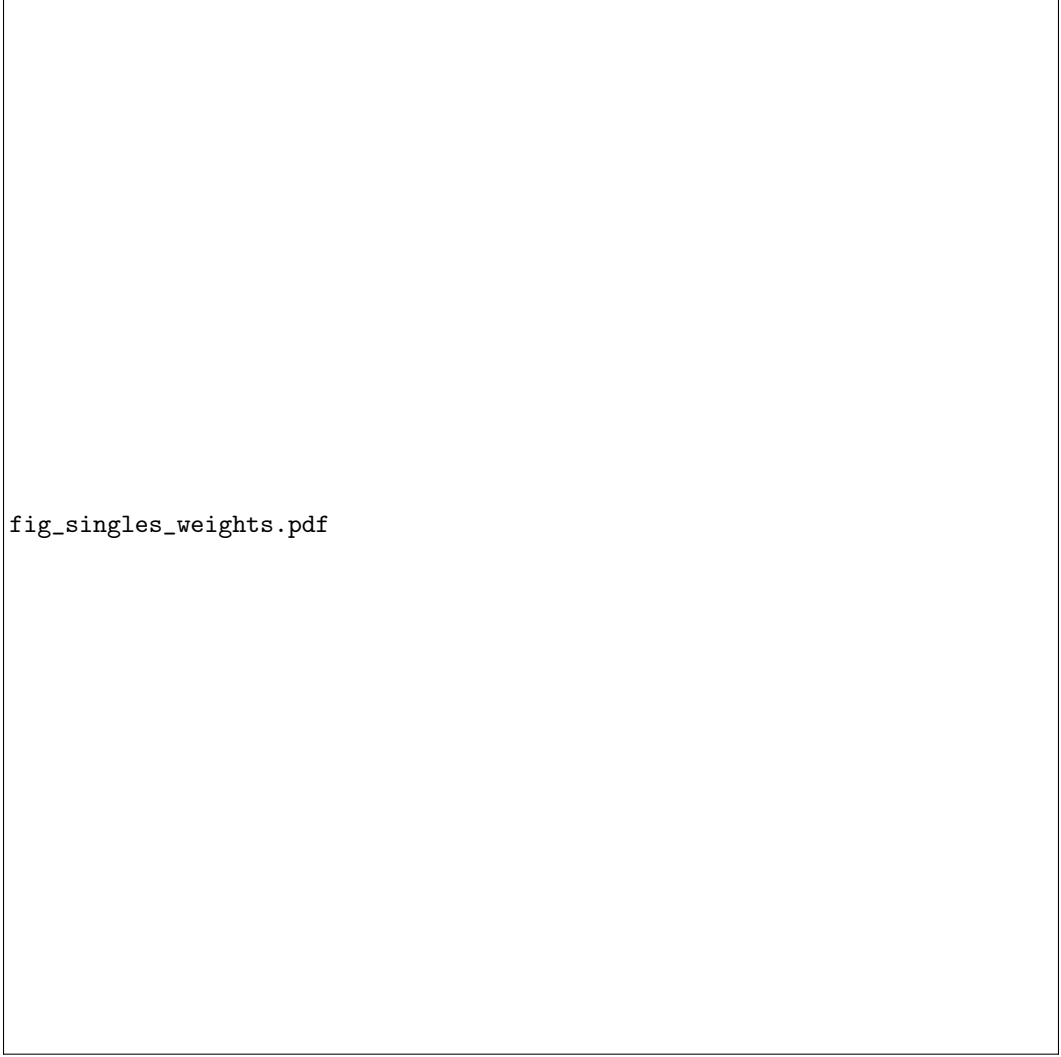
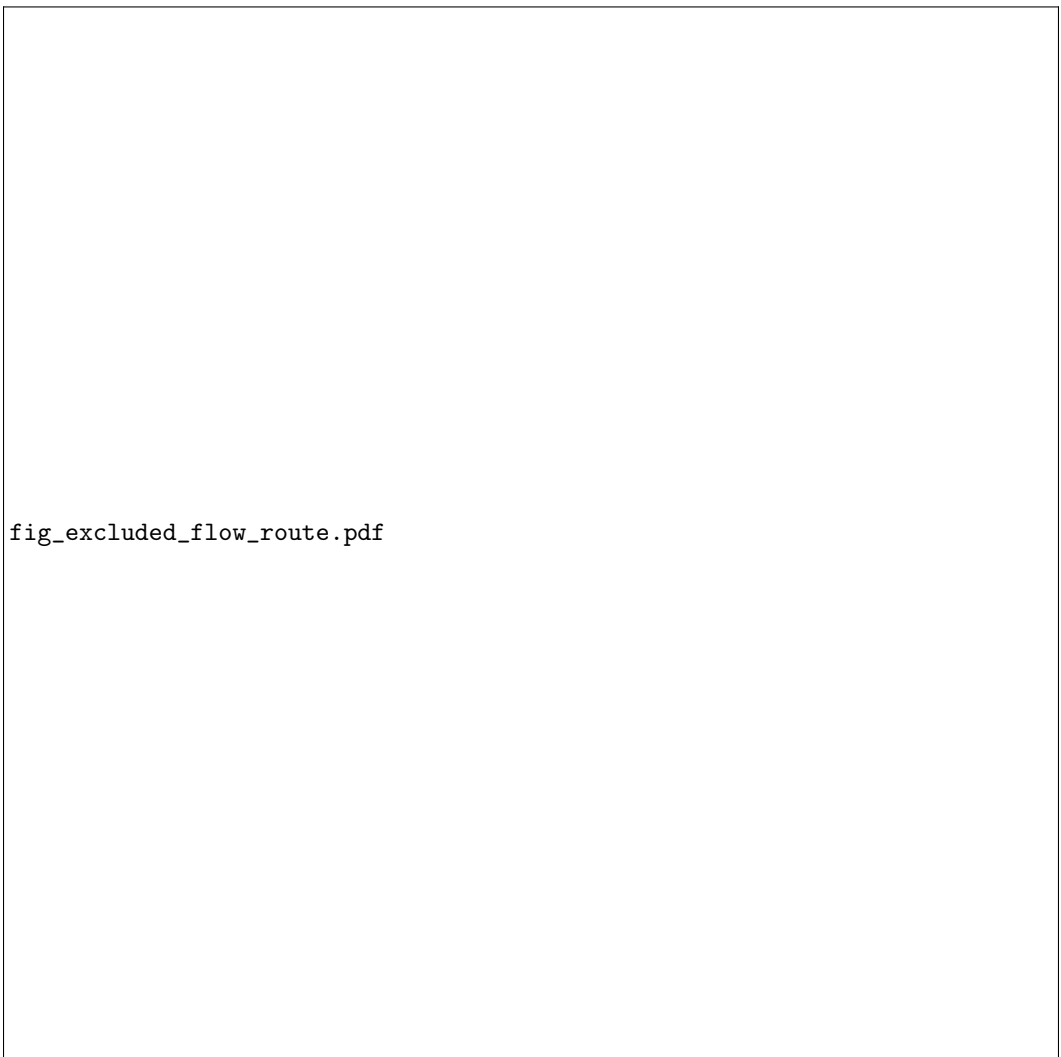


Figure 8: **The learned temperature never exceeds one.** Supports Appendix F. Left: $T_\theta(h)$ read from trained checkpoints, binned by the prior’s top-1 margin. The dashed line marks $T = 1$, above which the mechanism would soften the prior; learned values sit between 0.73 and 0.80 everywhere and are flat in the margin. Because the forward pass applies $\text{clamp}(T, \min=1)$, the effective temperature was 1 at every state and the mechanism was inactive throughout. Right: sampled-flip rate by margin for the same checkpoints, showing the decay above a margin of 8 shared by all architectures.



fig_singles_weights.pdf

Figure 9: **Per-component effect sizes and composition weights.** Supports Appendix G. Left: effect size of each component guide sampled alone, under a direct policy rather than a composition operator; the components differ, with one producing a negative mean shift. Right: effect size and uniqueness of the composed sampler under different weight vectors, from equal weights to mass concentrated on a single component. Concentrating weight changes the effect size but does not lift it out of the band reported in Section 5.

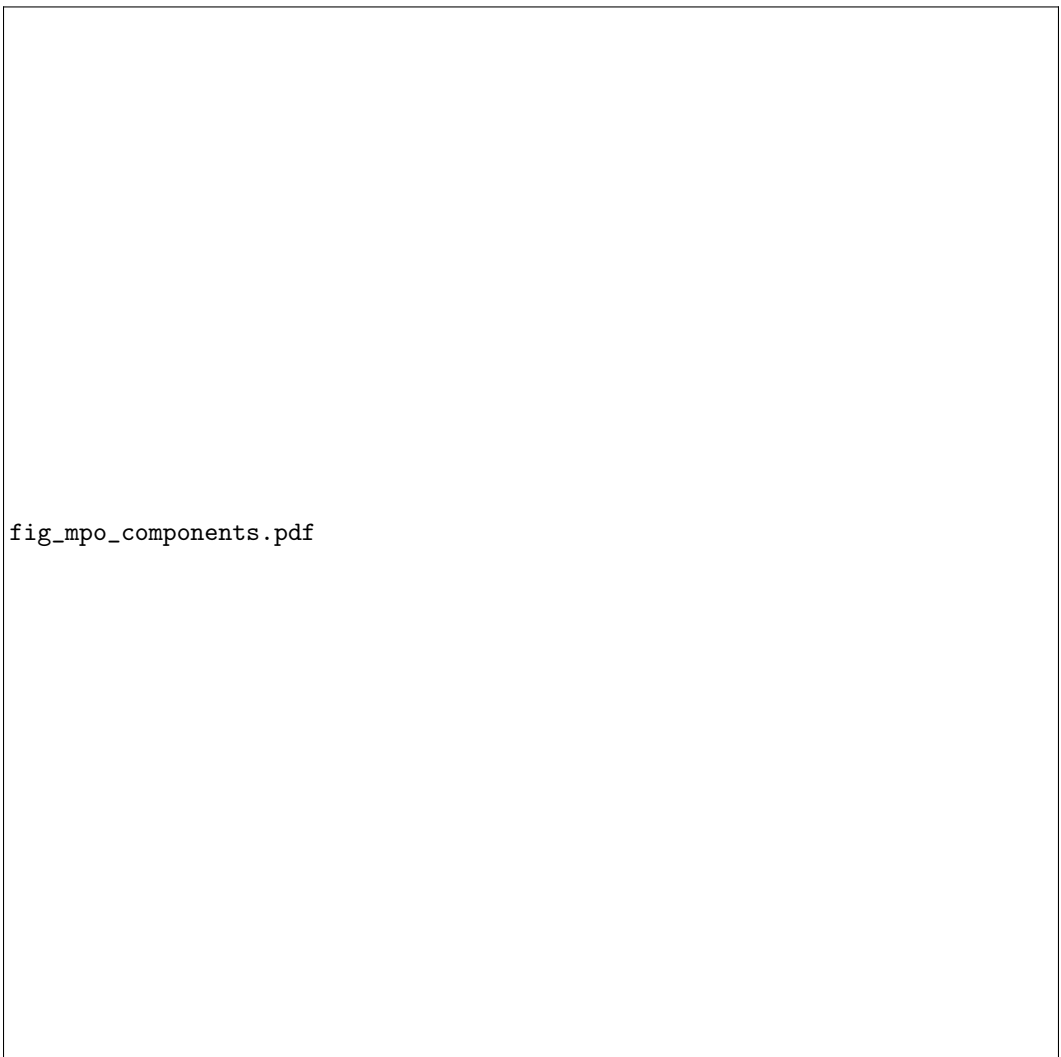


fig_excluded_flow_route.pdf

Figure 10: **Delivery failure in the excluded flow-route runs.** Supports Appendix G. Left: flip rate by margin, identically zero in every bin. Right: flip rate by sequence position, identically zero at every position. The corresponding rollout diagnostics report a residual norm of exactly 0.000 at every state, against 0.37 to 8.18 for the same checkpoints evaluated on the policy route. This is the signature of a guide that is computed but not applied, and is distinguishable from the bound reported in the main text precisely because delivery is measured separately from the flip rate.



Figure 11: **Distribution statistics over the course of training.** Supports Section 5. From left: 3D-to-SMILES conversion rate; share of the most frequent molecule within a window; effective number of distinct molecules, computed as the exponential of the Shannon entropy over a window; and mean heavy-atom count. All are computed over successive windows of the recorded generation stream. Conversion holds near 0.95 and the concentration measures are flat, so none of the fine-tuning runs collapses onto a small set of molecules. The unique-count fraction is not shown, as it saturates once a window exceeds the number of distinct molecules and is insensitive to concentration.



fig_mpo_components.pdf

Figure 12: **Per-component scores of the Osimertinib objective.** Supports Section 5. Mean score of each leaf scorer over the top-100 molecules of every fine-tuned configuration. Three of the four components are near-saturated under all configurations and one carries almost all of the variation. The profiles are close to identical across configurations, including one never trained on this objective, which indicates that the aggregate scores in Table 8 do not arise from any configuration having moved a particular axis.



Figure 13: **Top-10 mean against oracle calls.** Supports Appendix ???. Best-so-far top-10 score as a function of the number of molecules scored, for each fine-tuned configuration, with the GEOM best-of-10k baseline marked. The curve is the quantity integrated by the AUC metric of Equation (??). [FILL: GENERATE WITH HARVEST_EVAL.PY -EXTENDED -PLOTS_DIR; THE FIGURE IS PRODUCED BUT WAS NOT INCLUDED IN THE CURRENT SET.]



Figure 14: **Compound quality against benchmark score.** Supports Section 6. Quality-filter pass rate over the top-100 molecules against the top-10 benchmark score, one point per configuration, with the GEOM-Drugs reference marked. Configurations that score higher do not do so by producing molecules that fail the filters. [FILL: GENERATE WITH HARVEST_EVAL.PY -EXTENDED -PLOTS_DIR.]